

Tutorial 6

Bases and matrix operators

Homework

1. Question 2: Provide a detailed solution to Part 2.
2. Question 6: Provide detailed solutions to *all* parts.
3. Question 7: Provide a detailed solution to Part 2.
4. Question 9: Provide a detailed solution to Part 2.
5. Question 10: Provide a detailed solution to any one subpart of your choice.
6. Question 19: Provide a detailed solution to Part (b).
7. Question 20: Choose either Part (a) or Part (b) and provide a detailed solution to the selected part.
8. Question 23: Provide detailed solutions to *all* parts.
9. Question 28: Provide detailed solutions to *all* parts.

Question 1. Show that the following set of vectors forms a basis for $\mathbb{R}^2$ and $\mathbb{R}^3$ respectively.

1. $\{ (2, 1), (3, 0) \}$
2. $\{ (3, 1, -4), (2, 5, 6), (1, 4, 8) \}$

Question 2. Show that the following matrices form a basis for $\mathcal{M}_{2 \times 2}$.

1. $\begin{pmatrix} 3 & 6 \\ 3 & -6 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -8 \\ -12 & -4 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ -1 & 2 \end{pmatrix}$
2. $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$

Question 3. In each part, show that the set of vectors is not a basis for $\mathbb{R}^3$

1. $\{ (2, -3, 1), (4, 1, 1), (0, -7, 1) \}$
2. $\{ (1, 6, 4), (2, 4, -1), (-1, 2, 5) \}$

Question 4. Show that the following matrices do not form a basis for $\mathcal{M}_{2 \times 2}$.

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 2 & -2 \\ 3 & 2 \end{pmatrix}, \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$$

Question 5. Find the coordinate vector of $\mathbf{w}$ relative to the basis $S = \{ \mathbf{u}_1, \mathbf{u}_2 \}$ for $\mathbb{R}^2$.

1. $\mathbf{u}_1 = (2, -4), \mathbf{u}_2 = (3, 8), \mathbf{w} = (1, 1)$
2. $\mathbf{u}_1 = (1, 1), \mathbf{u}_2 = (0, 2), \mathbf{w} = (a, b)$
3. $\mathbf{u}_1 = (1, -1), \mathbf{u}_2 = (1, 1), \mathbf{w} = (1, 0)$
4. $\mathbf{u}_1 = (1, -1), \mathbf{u}_2 = (1, 1), \mathbf{w} = (0, 1)$

Question 6. Find the coordinate vector of $\mathbf{u}$ relative to the basis $S = \{ \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \}$ for $\mathbb{R}^3$

1. $\mathbf{v}_1 = (1, 0, 0), \mathbf{v}_2 = (2, 2, 0), \mathbf{v}_3 = (3, 3, 3), \mathbf{u} = (2, -1, 3)$
2. $\mathbf{v}_1 = (1, 2, 3), \mathbf{v}_2 = (-4, 5, 6), \mathbf{v}_3 = (7, -8, 9), \mathbf{u} = (5, -12, 3)$

Question 7. For each case, first show that the set $S = \{ A_1, A_2, A_3, A_4 \}$ is a basis for $\mathcal{M}_{2 \times 2}$, then express A as a linear combination of the matrices in S , and then find the coordinate vector of A relative to S .

1. $A_1 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, A_2 = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, A_3 = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}, A_4 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}; A = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$
2. $A_1 = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, A_2 = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, A_3 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, A_4 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}; A = \begin{pmatrix} 6 & 2 \\ 5 & 3 \end{pmatrix}$

Question 8. In each part, let $T_A : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be multiplication by A , and let $\{ \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3 \}$ be the standard basis for $\mathbb{R}^3$. Determine whether the set $\{ T_A(\mathbf{e}_1), T_A(\mathbf{e}_2), T_A(\mathbf{e}_3) \}$ is linearly independent in $\mathbb{R}^3$

1. $A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & -3 \\ -1 & 2 & 0 \end{pmatrix}$
2. $A = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ -1 & 2 & 1 \end{pmatrix}$

Question 9. In each part, let $T_A : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be multiplication by A , and let $\mathbf{u} = (1, -2, -1)$. Find the coordinate vector of $T_A(\mathbf{u})$ relative to the basis $S = \{ (1, 1, 0), (0, 1, 1), (1, 1, 1) \}$ for $\mathbb{R}^3$.

$$1. A = \begin{pmatrix} 2 & -1 & 0 \\ 1 & 1 & 1 \\ 0 & -1 & 2 \end{pmatrix}$$

$$2. A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

Question 10. Find a basis for the solution space of the homogeneous linear system, and find the dimension of that space

1.

$$\begin{aligned} x_1 + x_2 - x_3 &= 0 \\ -2x_1 - x_2 + 2x_3 &= 0 \\ -x_1 + x_3 &= 0 \end{aligned}$$

2.

$$\begin{aligned} 3x_1 + x_2 + x_3 + x_4 &= 0 \\ 5x_1 - x_2 + x_3 - x_4 &= 0 \end{aligned}$$

3.

$$\begin{aligned} 2x_1 + x_2 + 3x_3 &= 0 \\ x_1 + 5x_3 &= 0 \\ x_2 + x_3 &= 0 \end{aligned}$$

4.

$$\begin{aligned} x_1 - 3x_2 + x_3 &= 0 \\ 2x_1 - 6x_2 + 2x_3 &= 0 \\ 3x_1 - 9x_2 + 3x_3 &= 0 \end{aligned}$$

5.

$$\begin{aligned} x_1 - 4x_2 + 3x_3 - x_4 &= 0 \\ 2x_1 - 8x_2 + 6x_3 - 2x_4 &= 0 \end{aligned}$$

6.

$$\begin{aligned} x + y + z &= 0 \\ 3x + 2y - 2z &= 0 \\ 4x + 3y - z &= 0 \\ 6x + 5y + z &= 0 \end{aligned}$$

Question 11. In each part, find a basis for the given subspace of $\mathbb{R}^3$, and state its dimension

1. The plane $3x - 2y + 5z = 0$.
2. The plane $x - y = 0$.
3. The line $x = 2t, y = -t, z = 4t$.
4. All vectors of the form (a, b, c) , where $b = a + c$.

Question 12. In each part, find a basis for the given subspace of $\mathbb{R}^4$, and state its dimension.

1. All vectors of the form $(a, b, c, 0)$.
2. All vectors of the form (a, b, c, d) , where $d = a + b$ and $c = a - b$.
- c) All vectors of the form (a, b, c, d) , where $a = b = c = d$.

Question 13. Show that the matrices

$$M_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad M_3 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad M_4 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

form a basis for $\mathcal{M}_{2 \times 2}$.

Question 14. Find the dimension of each of the following vector spaces.

1. The vector space of all diagonal $n \times n$ matrices.
2. The vector space of all symmetric $n \times n$ matrices.
3. The vector space of all upper triangular $n \times n$ matrices.
4. The vector space of all lower triangular $n \times n$ matrices.

Question 15. Find a standard basis vector in $\mathbb{R}^3$ that can be added to the set $\{\mathbf{v}_1, \mathbf{v}_2\}$ to produce a basis for $\mathbb{R}^3$.

1. $\mathbf{v}_1 = (-1, 2, 3)$, $\mathbf{v}_2 = (1, 2, -2)$.
2. $\mathbf{v}_1 = (1, -1, 0)$, $\mathbf{v}_2 = (3, 1, -2)$.

Question 16. Find standard basis vectors for $\mathbb{R}^4$ that can be added to the set $\{\mathbf{v}_1, \mathbf{v}_2\}$ to produce a basis for $\mathbb{R}^4$.

$$\mathbf{v}_1 = (1, -4, 2, -3), \quad \mathbf{v}_2 = (-3, 8, -4, 6)$$

Question 17. The vectors $\mathbf{v}_1 = (1, -2, 3)$ and $\mathbf{v}_2 = (0, 5, -3)$ are linearly independent. Add one vector to $\{\mathbf{v}_1, \mathbf{v}_2\}$ to obtain a basis for $\mathbb{R}^3$.

Question 18. The vectors $\mathbf{v}_1 = (1, 0, 0, 0)$ and $\mathbf{v}_2 = (1, 1, 0, 0)$ are linearly independent. Enlarge the set $\{\mathbf{v}_1, \mathbf{v}_2\}$ to a basis for $\mathbb{R}^4$.

Question 19. Consider the bases $B_1 = \{\mathbf{u}_1, \mathbf{u}_2\}$ and $B_2 = \{\mathbf{v}_1, \mathbf{v}_2\}$ for $\mathbb{R}^2$, where

(a)

$$\mathbf{u}_1 = \begin{pmatrix} 2 \\ 2 \end{pmatrix}, \quad \mathbf{u}_2 = \begin{pmatrix} 4 \\ -1 \end{pmatrix}, \quad \mathbf{v}_1 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$$

(b)

$$\mathbf{u}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{u}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \mathbf{v}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$$

1. Find the transition matrix from B_2 to B_1
2. Find the transition matrix from B_1 to B_2
3. Compute the coordinate vector $[\mathbf{w}]_{B_1}$, where

$$\mathbf{w} = \begin{pmatrix} 3 \\ -5 \end{pmatrix}$$

and compute $[\mathbf{w}]_{B_2}$ using the transition matrix from B_1 to B_2

4. Compute $[\mathbf{w}]_{B_2}$ directly

Question 20. Consider the bases $B_1 = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ and $B_2 = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ for $\mathbb{R}^3$, where

(a)

$$\mathbf{u}_1 = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{u}_2 = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}, \quad \mathbf{u}_3 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \quad \mathbf{v}_1 = \begin{pmatrix} 3 \\ 1 \\ -5 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ -3 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$$

(b)

$$\mathbf{u}_1 = \begin{pmatrix} -3 \\ 0 \\ -3 \end{pmatrix}, \quad \mathbf{u}_2 = \begin{pmatrix} -3 \\ 2 \\ -1 \end{pmatrix}, \quad \mathbf{u}_3 = \begin{pmatrix} 1 \\ 6 \\ -1 \end{pmatrix}, \quad \mathbf{v}_1 = \begin{pmatrix} -6 \\ -6 \\ 0 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} -2 \\ -6 \\ 4 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} -2 \\ -3 \\ 7 \end{pmatrix}$$

1. Find the transition matrix from B_2 to B_1
2. Find the transition matrix from B_1 to B_2
3. Compute the coordinate vector $[\mathbf{w}]_{B_1}$, where

$$\mathbf{w} = \begin{pmatrix} -5 \\ 8 \\ -5 \end{pmatrix}$$

and compute $[\mathbf{w}]_{B_2}$ using the transition matrix from B_1 to B_2

4. Compute $[\mathbf{w}]_{B_2}$ directly

Question 21. Let $B_1 = \{\mathbf{u}_1, \mathbf{u}_2\}$ and $B_2 = \{\mathbf{v}_1, \mathbf{v}_2\}$ be the bases for $\mathbb{R}^2$, where

$$\mathbf{u}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \mathbf{u}_2 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}, \quad \mathbf{v}_1 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 4 \end{pmatrix}.$$

An efficient way to compute the transition matrix $P_{B_1 \rightarrow B_2}$ is as follows

Step 1. Form the matrix $(B_2 \mid B_1)$

Step 2. Use elementary row operations to reduce the matrix in Step 1 to reduced row echelon form

Step 3. The resulting matrix will be $(I \mid P_{B_1 \rightarrow B_2})$

Step 4. Extract the matrix $P_{B_1 \rightarrow B_2}$ from the right side of the matrix in Step 3

In diagram

$$(\text{new basis} \mid \text{old basis}) \xrightarrow{\text{row operations}} (I \mid \text{transition from old to new}) \quad (1)$$

1. Apply the above procedure to find the transition matrix $P_{B_2 \rightarrow B_1}$
2. Apply the above procedure to find the transition matrix $P_{B_1 \rightarrow B_2}$
3. Confirm that $P_{B_2 \rightarrow B_1}$ and $P_{B_1 \rightarrow B_2}$ are inverses of one another
4. Let $\mathbf{w} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Find $[\mathbf{w}]_{B_1}$ and then use the matrix $P_{B_1 \rightarrow B_2}$ to compute $[\mathbf{w}]_{B_2}$ from $[\mathbf{w}]_{B_1}$
5. Let $\mathbf{w} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$. Find $[\mathbf{w}]_{B_2}$ and then use the matrix $P_{B_2 \rightarrow B_1}$ to compute $[\mathbf{w}]_{B_1}$ from $[\mathbf{w}]_{B_2}$

Question 22. Let S be the standard basis for $\mathbb{R}^2$, and let $B = \{\mathbf{v}_1, \mathbf{v}_2\}$ be the basis in which

$$\mathbf{v}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$$

1. Find the transition matrix $P_{B \rightarrow S}$ by inspection
2. Use Formula (1) to find the transition matrix $P_{S \rightarrow B}$
3. Confirm that $P_{B \rightarrow S}$ and $P_{S \rightarrow B}$ are inverses of one another
4. Let $\mathbf{w} = \begin{pmatrix} 5 \\ -3 \end{pmatrix}$. Find $[\mathbf{w}]_B$ and then use the matrix $P_{B \rightarrow S}$ to compute $[\mathbf{w}]_S$ from $[\mathbf{w}]_B$
5. Let $\mathbf{w} = \begin{pmatrix} 3 \\ -5 \end{pmatrix}$. Find $[\mathbf{w}]_S$ and then use the matrix $P_{S \rightarrow B}$ to compute $[\mathbf{w}]_B$ from $[\mathbf{w}]_S$

Question 23. Let S be the standard basis for $\mathbb{R}^3$, and let $B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be the basis in which

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 2 \\ 5 \\ 0 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 3 \\ 3 \\ 8 \end{pmatrix}$$

1. Find the transition matrix $P_{B \rightarrow S}$ by inspection
2. Use Formula (1) to find the transition matrix $P_{S \rightarrow B}$
3. Confirm that $P_{B \rightarrow S}$ and $P_{S \rightarrow B}$ are inverses of one another

4. Let $\mathbf{w} = \begin{pmatrix} 5 \\ -3 \\ 1 \end{pmatrix}$. Find $[\mathbf{w}]_B$ and then use the matrix $P_{B \rightarrow S}$ to compute $[\mathbf{w}]_S$ from $[\mathbf{w}]_B$
5. Let $\mathbf{w} = \begin{pmatrix} 3 \\ -5 \\ 0 \end{pmatrix}$. Find $[\mathbf{w}]_S$ and then use the matrix $P_{S \rightarrow B}$ to compute $[\mathbf{w}]_B$ from $[\mathbf{w}]_S$

Question 24. Let $S = \{\mathbf{e}_1, \mathbf{e}_2\}$ be the standard basis for $\mathbb{R}^2$, and let $B = \{\mathbf{v}_1, \mathbf{v}_2\}$ be the basis that results when the vectors in S are reflected about the line $y = x$.

1. Find the transition matrix $P_{B \rightarrow S}$
2. Show that $P_{B \rightarrow S}^\top = P_{S \rightarrow B}$

Question 25. Let $S = \{\mathbf{e}_1, \mathbf{e}_2\}$ be the standard basis for $\mathbb{R}^2$, and let $B = \{\mathbf{v}_1, \mathbf{v}_2\}$ be the basis that results when the vectors in S are reflected about the line that makes an angle θ with the positive x -axis.

1. Find the transition matrix $P_{B \rightarrow S}$
2. Show that $P_{B \rightarrow S}^\top = P_{S \rightarrow B}$

Question 26. Find the domain and codomain of the transformation $T_A(\mathbf{x}) = A\mathbf{x}$

- | | |
|-------------------------------------|-------------------------------------|
| 1. $A \in \mathcal{M}_{3 \times 2}$ | 2. $A \in \mathcal{M}_{2 \times 3}$ |
| 3. $A \in \mathcal{M}_{3 \times 3}$ | 4. $A \in \mathcal{M}_{1 \times 6}$ |
| 5. $A \in \mathcal{M}_{4 \times 5}$ | 6. $A \in \mathcal{M}_{5 \times 4}$ |
| 7. $A \in \mathcal{M}_{4 \times 4}$ | 8. $A \in \mathcal{M}_{3 \times 1}$ |

Question 27. Find the domain and codomain of the transformation defined by the equations

1.

$$\begin{aligned} w_1 &= 4x_1 + 5x_2 \\ w_2 &= x_1 - 8x_2 \end{aligned}$$

2.

$$\begin{aligned} w_1 &= 5x_1 - 7x_2 \\ w_2 &= 6x_1 + x_2 \\ w_3 &= 2x_1 + 3x_2 \end{aligned}$$

3.

$$\begin{aligned} w_1 &= x_1 - 4x_2 + 8x_3 \\ w_2 &= -x_1 + 4x_2 + 2x_3 \\ w_3 &= -3x_1 + 2x_2 - 5x_3 \end{aligned}$$

4.

$$\begin{aligned} w_1 &= 2x_1 + 7x_2 - 4x_3 \\ w_2 &= 4x_1 - 3x_2 + 2x_3 \end{aligned}$$

Question 28. Find the standard matrix for the transformation defined below

1.

$$\begin{aligned} w_1 &= 2x_1 - 3x_2 + x_3 \\ w_2 &= 3x_1 + 5x_2 - x_3 \end{aligned}$$

2.

$$\begin{aligned} w_1 &= 7x_1 + 2x_2 - 8x_3 \\ w_2 &= -x_2 + 5x_3 \\ w_3 &= 4x_1 + 7x_2 - x_3 \end{aligned}$$

$$3. \quad T \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_2 \\ -x_1 \\ x_1 + 3x_2 \\ x_1 - x_2 \end{pmatrix}$$

$$4. \quad T \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 7x_1 + 2x_2 - x_3 + x_4 \\ x_2 + x_3 \\ -x_1 \end{pmatrix}$$

$$5. \quad T \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$6. \quad T \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} x_4 \\ x_1 \\ x_3 \\ x_2 \\ x_1 - x_3 \end{pmatrix}$$

Question 29. Find $T_A(\mathbf{x})$.

$$1. \quad A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \mathbf{x} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$$

$$2. \quad A = \begin{pmatrix} -1 & 2 & 0 \\ 3 & 1 & 5 \end{pmatrix}, \mathbf{x} = \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix}$$

$$3. \quad A = \begin{pmatrix} -2 & 1 & 4 \\ 3 & 5 & 7 \\ 6 & 0 & -1 \end{pmatrix}, \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$4. \quad A = \begin{pmatrix} -1 & 1 \\ 2 & 4 \\ 7 & 8 \end{pmatrix}, \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Question 30. The images of the standard basis vectors for $\mathbb{R}^3$ are given for a linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$. Find the standard matrix for the transformation, and find $T(\mathbf{x})$.

$$1. T(\mathbf{e}_1) = \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix}, T(\mathbf{e}_2) = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, T(\mathbf{e}_3) = \begin{pmatrix} 4 \\ -3 \\ -1 \end{pmatrix}, \mathbf{x} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$$

$$2. T(\mathbf{e}_1) = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}, T(\mathbf{e}_2) = \begin{pmatrix} -3 \\ -1 \\ 0 \end{pmatrix}, T(\mathbf{e}_3) = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \mathbf{x} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$$

Question 31. Use matrix multiplication to find the reflection of $(-1, 2)$ about the

1. x -axis
2. y -axis
3. line $y = x$

Question 32. Use matrix multiplication to find the reflection of (a, b) about the

1. x -axis
2. y -axis
3. line $y = x$

Question 33. Use matrix multiplication to find the reflection of $(2, -5, 3)$ about the

1. xy -plane
2. xz -plane
3. yz -plane

Question 34. Use matrix multiplication to find the reflection of $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ about the

1. xy -plane
2. xz -plane
3. yz -plane

Question 35. Use matrix multiplication to find the orthogonal projection of $\begin{pmatrix} 2 \\ -5 \end{pmatrix}$ onto the

1. x -axis
2. y -axis

Question 36. Use matrix multiplication to find the orthogonal projection of $\begin{pmatrix} a \\ b \end{pmatrix}$ onto the

1. x -axis2. y -axis

Question 37. Use matrix multiplication to find the orthogonal projection of $\begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix}$ onto the

1. xy -plane2. xz -plane3. yz -plane

Question 38. Use matrix multiplication to find the orthogonal projection of $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ onto the

1. xy -plane2. xz -plane3. yz -plane

Question 39. Use matrix multiplication to find the image of the vector $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$ when it is rotated about the origin through an angle of

1. $\theta = 30^\circ$ 2. $\theta = -60^\circ$ 3. $\theta = 45^\circ$ 4. $\theta = 90^\circ$

Question 40. Use matrix multiplication to find the image of the nonzero vector $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ when it is rotated about the origin through

1. a positive angle θ 2. a negative angle $-\theta$

Question 41. Use matrix multiplication to find the image of the vector $\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$ if it is rotated

1. 30° clockwise about the positive x -axis.2. 30° counterclockwise about the positive y -axis.3. 45° clockwise about the positive y -axis.4. 90° counterclockwise about the positive z -axis.

Question 42. Use matrix multiplication to find:

1. The contraction of $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$ with factor $\alpha = \frac{1}{2}$.
2. The dilation of $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$ with factor $\alpha = 3$.

Question 43. Use matrix multiplication to find:

1. The contraction of $\begin{pmatrix} a \\ b \end{pmatrix}$ with factor $\frac{1}{\alpha}$, where $\alpha > 1$.
2. The dilation of $\begin{pmatrix} a \\ b \end{pmatrix}$ with factor α , where $\alpha > 1$.

Question 44. Use matrix multiplication to find:

1. The contraction of $\begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$ with factor $\frac{1}{4}$.
2. The dilation of $\begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$ with factor 2.

Question 45. Use matrix multiplication to find:

1. The contraction of $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ with factor $\frac{1}{\alpha}$, where $\alpha > 1$.
2. The dilation of $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ with factor α , where $\alpha > 1$.

Question 46. Use matrix multiplication to find:

1. The compression of $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$ in the x -direction with factor $\frac{1}{2}$.
2. The compression of $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$ in the y -direction with factor $\frac{1}{2}$.

Question 47. Use matrix multiplication to find:

1. The expansion of $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$ in the x -direction with factor 3.
2. The expansion of $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$ in the y -direction with factor 3.

Question 48. Use matrix multiplication to find:

1. The compression of $\begin{pmatrix} a \\ b \end{pmatrix}$ in the x -direction with factor $\frac{1}{\alpha}$, where $\alpha > 1$.
2. The expansion of $\begin{pmatrix} a \\ b \end{pmatrix}$ in the y -direction with factor α , where $\alpha > 1$.

Question 49. In each part, determine whether the operators T_1 and T_2 commute, i.e. whether $T_1 \circ T_2 = T_2 \circ T_1$.

1. $T_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is the reflection about the line $y = x$, and $T_2 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is the orthogonal projection onto the x -axis.
2. $T_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is the reflection about the x -axis, and $T_2 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is the reflection about the line $y = x$.
3. $T_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is the orthogonal projection onto the x -axis, and $T_2 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is the orthogonal projection onto the y -axis.
4. $T_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is the rotation about the origin through an angle of $\frac{\pi}{4}$, and $T_2 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is the reflection about the y -axis.
5. $T_1 : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is a dilation with factor α , and $T_2 : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is a contraction with factor $\frac{1}{\alpha}$, where $\alpha > 1$.
6. $T_1 : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is the rotation about the x -axis through an angle θ_1 , and $T_2 : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is the rotation about the z -axis through an angle θ_2 .

Question 50. Find the standard matrix for the stated composition in $\mathbb{R}^2$.

1. A rotation of 90° , followed by a reflection about the line $y = x$.
2. An orthogonal projection onto the y -axis, followed by a contraction with factor $\frac{1}{2}$.
3. A reflection about the x -axis, followed by a dilation with factor 3, followed by a rotation about the origin of 60° .
4. A rotation about the origin of 60° , followed by an orthogonal projection onto the x -axis, followed by a reflection about the line $y = x$.

5. A dilation with factor 2, followed by a rotation about the origin of 45° , followed by a reflection about the y -axis.
6. A rotation about the origin of 15° , followed by a rotation about the origin of 105° , followed by a rotation about the origin of 60° .

Question 51. Find the standard matrix for the stated composition in $\mathbb{R}^3$.

1. A reflection about the yz -plane, followed by an orthogonal projection onto the xz -plane.
2. A rotation of 45° about the y -axis, followed by a dilation with factor $\sqrt{2}$.
3. An orthogonal projection onto the xy -plane, followed by a reflection about the yz -plane.
4. A rotation of 30° about the x -axis, followed by a rotation of 30° about the z -axis, followed by a contraction with factor $\frac{1}{4}$.
5. A reflection about the xy -plane, followed by a reflection about the xz -plane, followed by an orthogonal projection onto the yz -plane.
6. A rotation of 270° about the x -axis, followed by a rotation of 90° about the y -axis, followed by a rotation of 180° about the z -axis.

Question 52. Let $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ be a vector in $\mathbb{R}^2$. Consider the linear transformations $T_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and $T_2 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$T_1(\mathbf{x}) = \begin{pmatrix} x_1 + x_2 \\ x_1 - x_2 \end{pmatrix}, \quad T_2(\mathbf{x}) = \begin{pmatrix} 3x_1 \\ 2x_1 + 4x_2 \end{pmatrix}$$

1. Find the standard matrices for T_1 and T_2 .
2. Find the standard matrices for $T_1 \circ T_2$ and $T_2 \circ T_1$.
3. Find the standard matrices for $T_1 \circ T_2 \circ T_1$ and $T_1 \circ T_2 \circ T_2$.
4. Use the matrices obtained in part 2 to find formulas for $T_1(T_2(\mathbf{x}))$ and $T_2(T_1(\mathbf{x}))$

Question 53. Let $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$ be a vector in $\mathbb{R}^3$. Consider the linear transformations $T_1 : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ and $T_2 : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by

$$T_1(\mathbf{x}) = \begin{pmatrix} 4x_1 \\ -2x_1 + x_2 \\ -x_1 - 3x_3 \end{pmatrix}, \quad T_2(\mathbf{x}) = \begin{pmatrix} x_1 + 2x_2 \\ 2x_3 \\ 4x_1 - x_3 \end{pmatrix}$$

1. Find the standard matrices for T_1 and T_2 .

2. Find the standard matrices for $T_1 \circ T_2$ and $T_2 \circ T_1$.
3. Find the standard matrices for $T_1 \circ T_2 \circ T_1$ and $T_1 \circ T_2 \circ T_2$.
4. Use the matrices obtained in part 2 to find formulas for $T_1(T_2(\mathbf{x}))$ and $T_2(T_1(\mathbf{x}))$